E-Commerce Product Data Analysis Report

Exploratory data analysis uncovering pricing strategies, discount patterns, and customer engagement insights from e-commerce product data.

Project Objective

This exploratory data analysis (EDA) examines an e-commerce product dataset to understand product pricing strategies, discount patterns, customer ratings, and review engagement. The goal is to uncover actionable insights that help businesses optimize pricing strategies, improve customer satisfaction, and identify high-performing products.

E-commerce platforms generate vast volumes of product and customer interaction data. By analyzing this data, businesses can identify trends in customer preferences, evaluate the effectiveness of discount strategies, and make data-driven decisions regarding product promotion and inventory planning.

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Dataset Overview

The dataset contains product information from an e-commerce platform. Each record represents a product with its pricing, rating, and review information.

Product Name

Name of the product listed on the platform

Category

Category to which the product belongs

Actual Price

Original price before discount

Discounted Price

Price after applying discount

Discount %

Percentage reduction from original price

Rating

Average customer rating

Rating Count

Total number of customer reviews

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Data Cleaning and Preparation

01

Remove Currency Symbols

Currency symbols and formatting characters were removed from price columns to enable conversion into numerical values.

02

Clean Discount Column

Percentage symbols were removed from the discount percentage column and converted into numeric format.

03

Handle Missing Values

Missing values were checked and addressed to ensure data completeness for analysis.

04

Correct Data Types

Data types of price, discount, rating, and review count columns were standardized for statistical analysis.

05

Create Savings Feature

A new feature representing the difference between actual and discounted price was calculated to understand customer savings.

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Price Distribution Analysis

Key Findings

The distribution of product prices reveals that most products fall within the low to mid-price range. Only a small number of products belong to the premium price segment. This indicates that the platform primarily targets customers looking for affordable products.

The pricing strategy appears to focus on accessibility, with the majority of offerings positioned in the value-conscious market segment.

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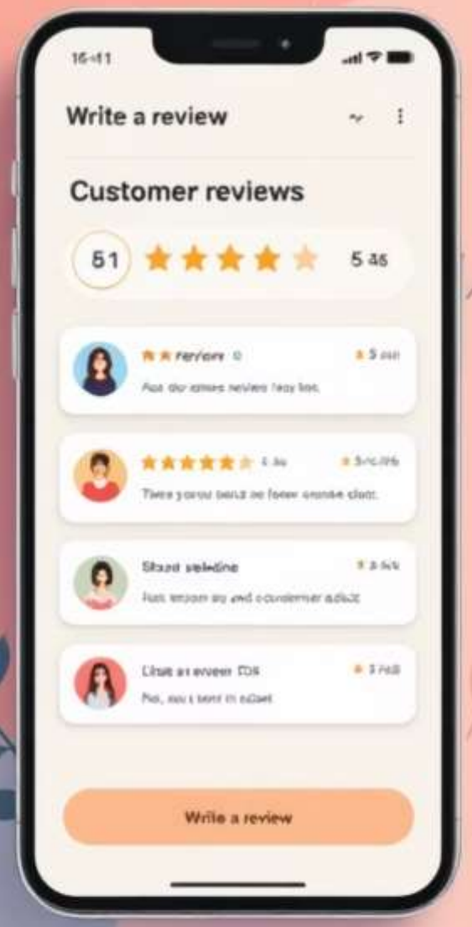

Discount and Rating Patterns

Discount Analysis

Many products offer moderate discounts. High discount percentages are typically associated with products that aim to increase visibility and attract customer engagement. Strategic discounting appears to be used as a promotional tool.

Rating Analysis

Customer ratings are generally concentrated between 4.0 and 4.5, indicating that most products receive positive feedback. Very few products have extremely low ratings, suggesting overall product quality is maintained across the platform.

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Customer Engagement Insights

Review Volume Variation

The number of reviews varies significantly across products. Some products have a very large number of reviews, indicating high popularity and customer engagement, while others have limited visibility.

Category Differences

Product categories differ significantly in terms of the number of listings, average price, and customer engagement. Some categories dominate the platform with a higher number of products and reviews.

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Key Insights Discovered

1. **Affordable Pricing Focus**

   A majority of products are priced within the affordable range, indicating the platform focuses on price-sensitive customers.

2. **Discount-Engagement Correlation**

   Products with higher discounts tend to receive more customer reviews and engagement.

3. **Visibility Opportunity**

   Highly rated products do not always receive a large number of reviews, indicating potential opportunities for better product promotion.

4. **Premium Product Challenge**

   Premium products generally have lower review counts compared to mid-range products.

5. **Category Performance**

   Certain categories generate significantly more customer interaction than others.

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Business Recommendations

Promote Highly Rated Products

Target highly rated products with low visibility to increase their sales potential through strategic placement and marketing campaigns.

Implement Targeted Discounts

Apply strategic discount strategies for products that are currently underperforming to boost engagement and sales.

Focus on High-Engagement Categories

Concentrate marketing efforts on categories with proven high customer engagement to maximize return on investment.

Encourage Customer Reviews

Implement initiatives to encourage more customer reviews, improving product credibility and visibility across the platform.

Conclusion

This project demonstrates how exploratory data analysis can be used to gain valuable insights from e-commerce product data. By analyzing pricing patterns, discount strategies, customer ratings, and review engagement, businesses can better understand product performance and customer behavior.

The findings from this analysis support decision-making in product promotion, pricing optimization, and category management. With further enhancements such as predictive analytics and recommendation systems, the analysis can be extended to provide even deeper insights for e-commerce platforms.

Next Steps: Implement predictive modeling to forecast product performance and integrate real-time analytics for continuous optimization.

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